

Goal: Increase Profits by 10% this year

To address the problem that Natalie wants to address, we would have to look into which sub-categories are selling the most. We would be able to increase profits by selling more of the best-selling items or reducing the worst-selling items. Another solution that is possible is to promote more in certain regions or sectors.

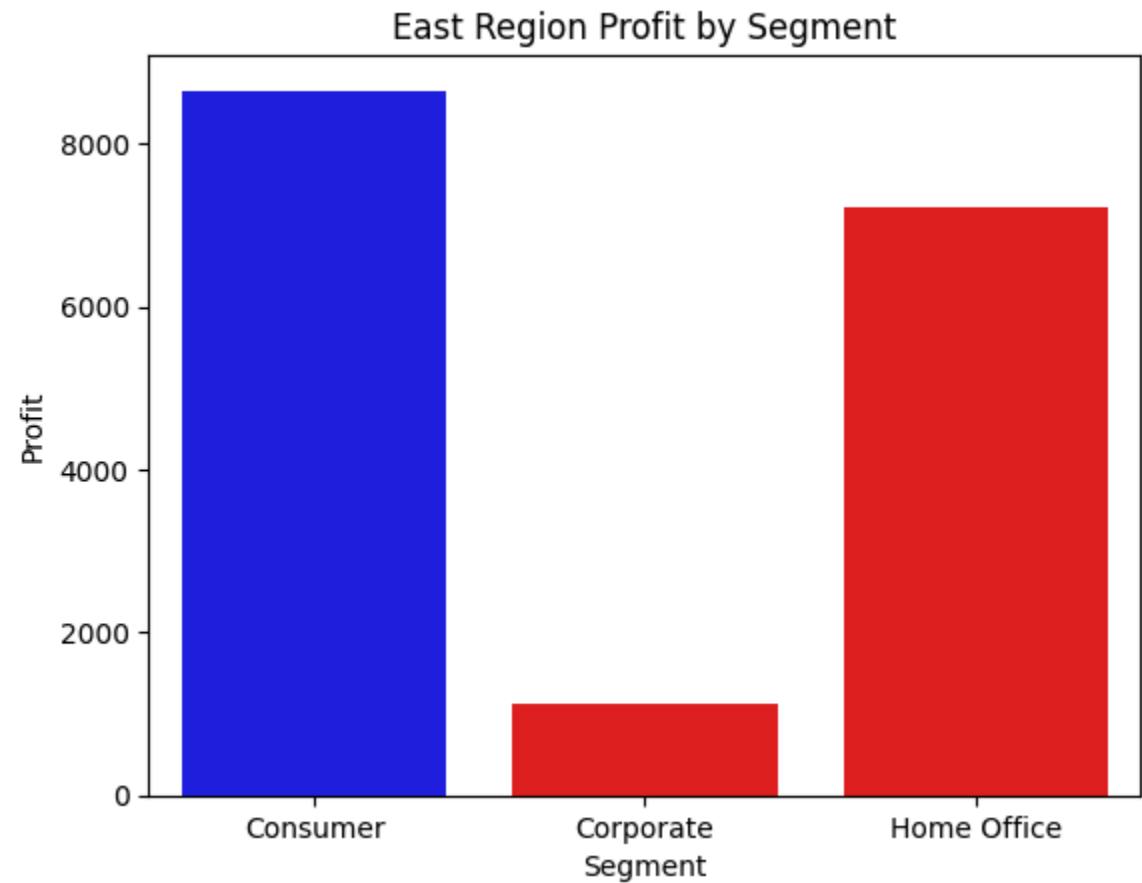
	Profit	Sales
Sub-Category		
Copiers	55617.8249	149528.0300
Phones	44515.7306	330007.0540
Accessories	41936.6357	167380.3180
Paper	34053.5693	78479.2060
Binders	30221.7633	203412.7330
Chairs	26590.1663	328449.1030
Storage	21278.8264	223843.6080
Appliances	18138.0054	107532.1610
Furnishings	13059.1436	91705.1640
Envelopes	6964.1767	16476.4020
Art	6527.7870	27118.7920
Labels	5546.2540	12486.3120
Machines	3384.7569	189238.6310
Fasteners	949.5182	3024.2800
Supplies	-1189.0995	46673.5380
Bookcases	-3472.5560	114879.9963
Tables	-17725.4811	206965.5320

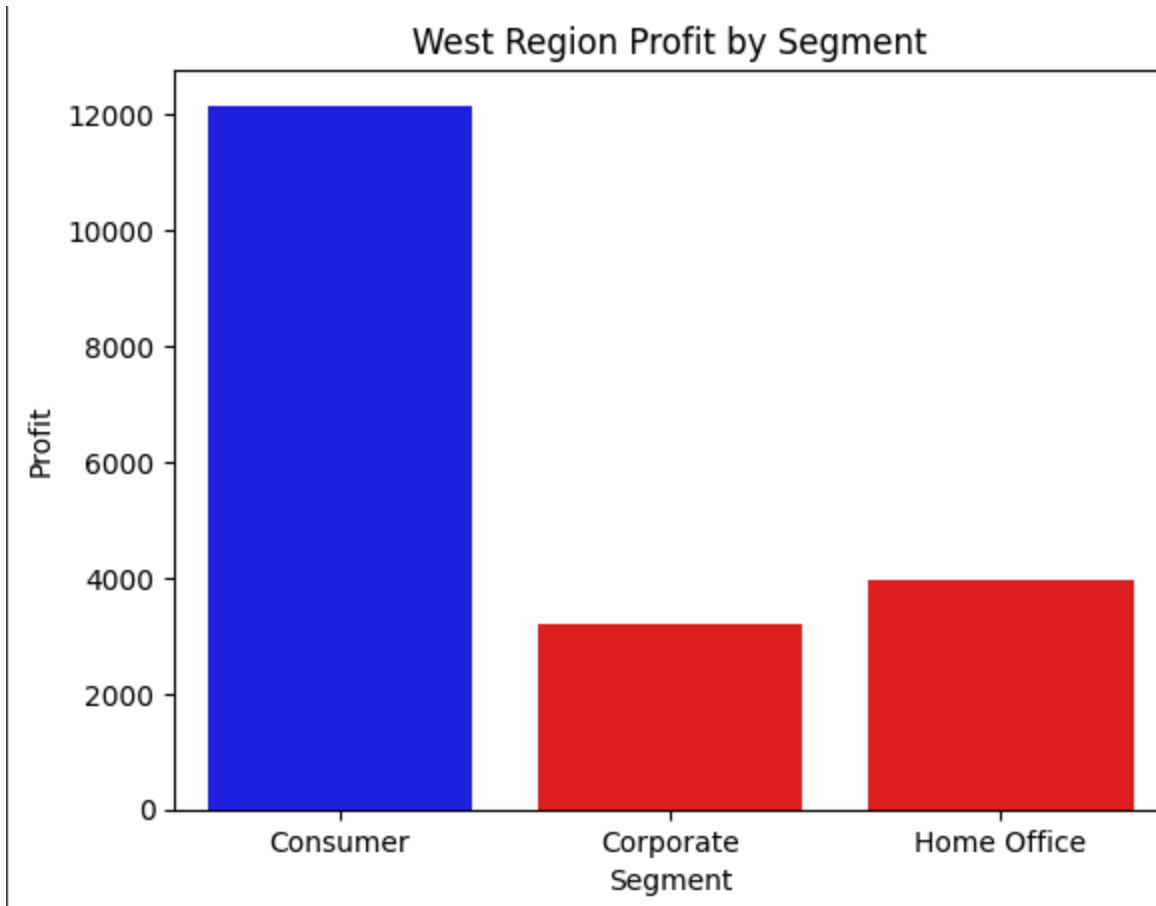
As we can see from Figure one, we see that Copiers are the first sub-category and make the most profit with the profit being \$55617.8249. We can clearly see that Copiers were beating the second sub-category by a huge margin which is about \$11,000. As a result, I would dive more into why Copiers are profiting way more compared to other Sub-Categories and which regions and the product names which are selling the most. I would also look into the discounts that we are giving in the selling of the copiers.

	Profit	Sales
Region		
South	3658.9067	9299.756
Central	15608.8413	37259.570
East	17022.8418	53219.462
West	19327.2351	49749.242

Firstly, I looked out the regions that were profiting the most. Referring to Figure 2, I found out that West, East, and Central were close to each other. I, then, chose the East and West region to explore more regarding which sectors they were selling more and which product name they were selling more compared to the rest. The figures three and four show that the east region was making

a profit of \$8, 567.93 in the consumer sectors while the West region was making \$12,158.87 in the consumer sector.





After figuring out that the West was profiting more in the consumer sector by selling copiers. I wanted to see which product they were selling the most in that region. The product name "Canon image CLASS 2200 Advanced Copier" stood out. Referred in figure five, we see that the product makes a sale of around 14,000 units and the profit is about \$6,719.98. Upon finding out that the Canon 2200 Advanced Copier sold more compared to the other copiers, I wanted to figure out whether it was the discount that is enabling them to have a high profit margin and how its discount compared to the other copiers.

Product Name	Profit	Sales
Hewlett Packard 310 Color Digital Copier	209.9930	1259.958
Brother DCP1000 Digital 3 in 1 Multifunction Ma...	224.9925	1199.960
Hewlett Packard LaserJet 3310 Copier	335.9944	959.984
Canon PC-428 Personal Copier	377.9811	1119.944
Hewlett Packard 610 Color Digital Copier / Printer	449.9910	999.980
Canon PC1080F Personal Copier	701.9883	1799.970
Sharp AL-1530CS Digital Copier	869.9826	2399.952
Canon PC1060 Personal Laser Copier	2267.9676	6719.904
Canon imageCLASS 2200 Advanced Copier	6719.9808	13999.960

Product Name	Discount	Profit
Canon PC1060 Personal Laser Copier	0.200000	755.989200
Brother DCP1000 Digital 3 in 1 Multifunction Ma...	0.200000	224.992500
Canon PC-428 Personal Copier	0.200000	188.990550
Hewlett Packard LaserJet 3310 Copier	0.200000	335.994400
Sharp AL-1530CS Digital Copier	0.200000	434.991300
Hewlett Packard 310 Color Digital Copier	0.133333	69.997667
Canon PC1080F Personal Copier	0.000000	701.988300
Canon imageCLASS 2200 Advanced Copier	0.000000	6719.980800
Hewlett Packard 610 Color Digital Copier / Printer	0.000000	449.991000

Surprisingly, I discovered the Canon 2200 Advanced Copier did not even offer a discount compared to the other copiers. It was also surprising to find out that the 2200 Advanced Copiers were more than double the sales of the second place product, which was the Personal Laser Copiers. My initial thought was that the data might be skewed to one year because it is possible that the 2200 Advanced Copiers were popular in one year. I found out the Canon 2200 Advanced Copiers were sold mostly in 2016 and 2017 and were not found in the years 2018 and 2019. Referring to the figure seven, I found out that it profited around \$15,680 in 2017 and \$9,520 in 2016.

		Profit	Discount
Year	Product Name		
2016	Brother DCP1000 Digital 3 in 1 Multifunction Ma...	89.9970	0.200000
	Canon Imageclass D680 Copier / Fax	69.9990	0.400000
	Canon PC-428 Personal Copier	913.9543	0.100000
	Canon PC1060 Personal Laser Copier	1889.9730	0.200000
	Canon imageCLASS 2200 Advanced Copier	9519.9728	0.200000
	Hewlett Packard 310 Color Digital Copier	569.9810	0.100000
	Hewlett Packard 610 Color Digital Copier / Printer	1724.9655	0.066667
	Hewlett Packard LaserJet 3310 Copier	2783.9536	0.200000
	Sharp AL-1530CS Digital Copier	179.9964	0.400000
2017	Brother DCP1000 Digital 3 in 1 Multifunction Ma...	89.9970	0.200000
	Canon Image Class D660 Copier	1691.9718	0.100000
	Canon Imageclass D680 Copier / Fax	1679.9760	0.100000
	Canon PC-428 Personal Copier	177.9911	0.300000
	Canon PC1080F Personal Copier	1037.9827	0.100000
	Canon imageCLASS 2200 Advanced Copier	15679.9552	0.066667
	Hewlett Packard 310 Color Digital Copier	209.9930	0.133333
	Hewlett Packard 610 Color Digital Copier / Printer	74.9985	0.400000
	Hewlett Packard LaserJet 3310 Copier	3623.9396	0.150000
	Sharp 1540cs Digital Laser Copier	329.9940	0.200000
	Sharp AL-1530CS Digital Copier	434.9913	0.200000
2018	Brother DCP1000 Digital 3 in 1 Multifunction Ma...	224.9925	0.200000
	Canon Imageclass D680 Copier / Fax	174.9975	0.200000
	Canon PC1080F Personal Copier	701.9883	0.000000
	Canon PC940 Copier	125.9972	0.400000
	Hewlett Packard 610 Color Digital Copier / Printer	624.9875	0.200000
	Hewlett Packard LaserJet 3310 Copier	239.9960	0.400000
	Sharp 1540cs Digital Laser Copier	384.9930	0.200000
	Sharp AL-1530CS Digital Copier	434.9913	0.200000
2019	Brother DCP1000 Digital 3 in 1 Multifunction Ma...	749.9750	0.066667
	Canon Imageclass D680 Copier / Fax	874.9875	0.200000
	Canon PC-428 Personal Copier	377.9811	0.200000
	Canon PC1060 Personal Laser Copier	2680.9617	0.100000
	Canon PC1080F Personal Copier	233.9961	0.000000
	Canon PC940 Copier	1966.4563	0.100000
	Hewlett Packard 310 Color Digital Copier	59.9980	0.200000
	Hewlett Packard 610 Color Digital Copier / Printer	699.9860	0.100000
	Hewlett Packard LaserJet 3310 Copier	335.9944	0.200000
	Sharp 1540cs Digital Laser Copier	934.9830	0.150000
	Sharp AL-1530CS Digital Copier	1014.9797	0.200000

Recommendation

The data analytics of this table made me question whether the Canon 2200 Advanced copiers stopped producing or whether these regions stopped selling. I would recommend Natalie to concentrate on bringing the Canon 2200 Advanced copiers back to the West region since it is the region where the copiers are popular. In addition, we

would be able to give a discount of 6%-20% for the copier looking at the trend from 2016 and 2017. I recommend focusing on the marketing and advertising of the Canon 2200 Advance copiers in the West and possibly lowering the discount or taking out the Hewlett Packard 310 Copier since we are barely profiting from this copier.